Source-Based AI Recommendation Explanations — Measurement v1

Response Scale

Unless otherwise indicated, participants will respond using a 7-point Likert scale:

1 = Strongly disagree
2 = Disagree
3 = Somewhat disagree
4 = Neither agree nor disagree
5 = Somewhat agree
6 = Agree
7 = Strongly agree

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1. Perceived Relevance

Construct Definition

The extent to which the participant perceives that the recommendation fits their stated needs and preferences.

Items

PR1. The recommended product is relevant to my needs.

PR2. The recommendation matches the preferences I provided.

PR3. The recommended product seems well suited to what I am looking for.

Source Basis

Adapted from recommender-system research on perceived recommendation quality, preference fit, and perceived personalization.

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2. Perceived Source Credibility

Construct Definition

The extent to which the participant perceives the information underlying the recommendation as credible, trustworthy, and reliable.

Items

SC1. The information supporting this recommendation seems credible.

SC2. The basis of this recommendation seems trustworthy.

SC3. The information used to generate this recommendation appears reliable.

Source Basis

Adapted from source-credibility research emphasizing perceived trustworthiness and expertise/reliability of an information source.

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3. Situational Privacy Concern

Construct Definition

The extent to which the recommendation makes the participant concerned or uncomfortable about the collection and use of their personal or behavioral information.

Items

PC1. This recommendation makes me concerned about how my personal information may have been used.

PC2. I feel uneasy about the possibility that my online activity was used to generate this recommendation.

PC3. I am concerned about how the AI shopping assistant may collect or use information about me.

Source Basis

Adapted from information-privacy concern research, particularly concerns surrounding collection, awareness, and use of personal information.

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4. Trust in the AI Recommendation

Construct Definition

The extent to which the participant believes that the AI shopping assistant and its recommendation are dependable and competent.

Items

TR1. I consider the AI shopping assistant's recommendation trustworthy.

TR2. I believe the AI shopping assistant provides reliable recommendations.

TR3. I have confidence in the recommendation provided by the AI shopping assistant.

TR4. I believe the AI shopping assistant is competent in making this recommendation.

Source Basis

Adapted from established e-commerce trust and recommendation-agent trust research.

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5. Reliance Intention

Construct Definition

The participant's willingness to use or follow the AI recommendation when making a purchase decision.

Items

RI1. I would rely on this recommendation when making my purchase decision.

RI2. I would be willing to follow the AI shopping assistant's recommendation.

RI3. I would use this AI recommendation as an important input in deciding which product to purchase.

Source Basis

Adapted from recommendation-agent adoption, decision-aid use, and willingness-to-depend measures.

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6. Behavioral Reliance

After completing the rating scales, participants will make a behavioral choice.

Decision Question

Based on the information you reviewed, what would you most likely do?

A. Choose the AeroSound X5 recommended by the AI

B. Continue searching for another pair of headphones

Coding

Accept recommendation = 1
Continue searching = 0

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7. Manipulation Checks

The previously developed manipulation-check items should be presented after the main dependent measures.

MC1. The recommendation was based on my previous browsing or shopping behavior.

MC2. The recommendation was based on information from other consumers' reviews.

MC3. The recommendation was based on the product's features and attributes.

MC4. The recommendation combined multiple sources of information.

MC5. The explanation clearly indicated what information source or sources were used to generate the recommendation.

Recommended Questionnaire Order

1. Experimental stimulus
2. Perceived Relevance
3. Perceived Source Credibility
4. Situational Privacy Concern
5. Trust
6. Reliance Intention
7. Behavioral Reliance
8. Manipulation Checks
9. Demographics
10. Debrief

Items within each multi-item construct should be randomized when the survey is implemented.